

1. Project Overview

This project involves a comprehensive **Business Intelligence report** developed for **Maven Market**, a multi-national retail chain operating across the **USA, Canada, and Mexico**. The report analyzes sales performance, profitability, customer demographics, and product return behavior to support strategic decision-making.

Goal: Empower stakeholders with data-driven insights to optimize sales strategies, identify high-value customers, and reduce product return rates.

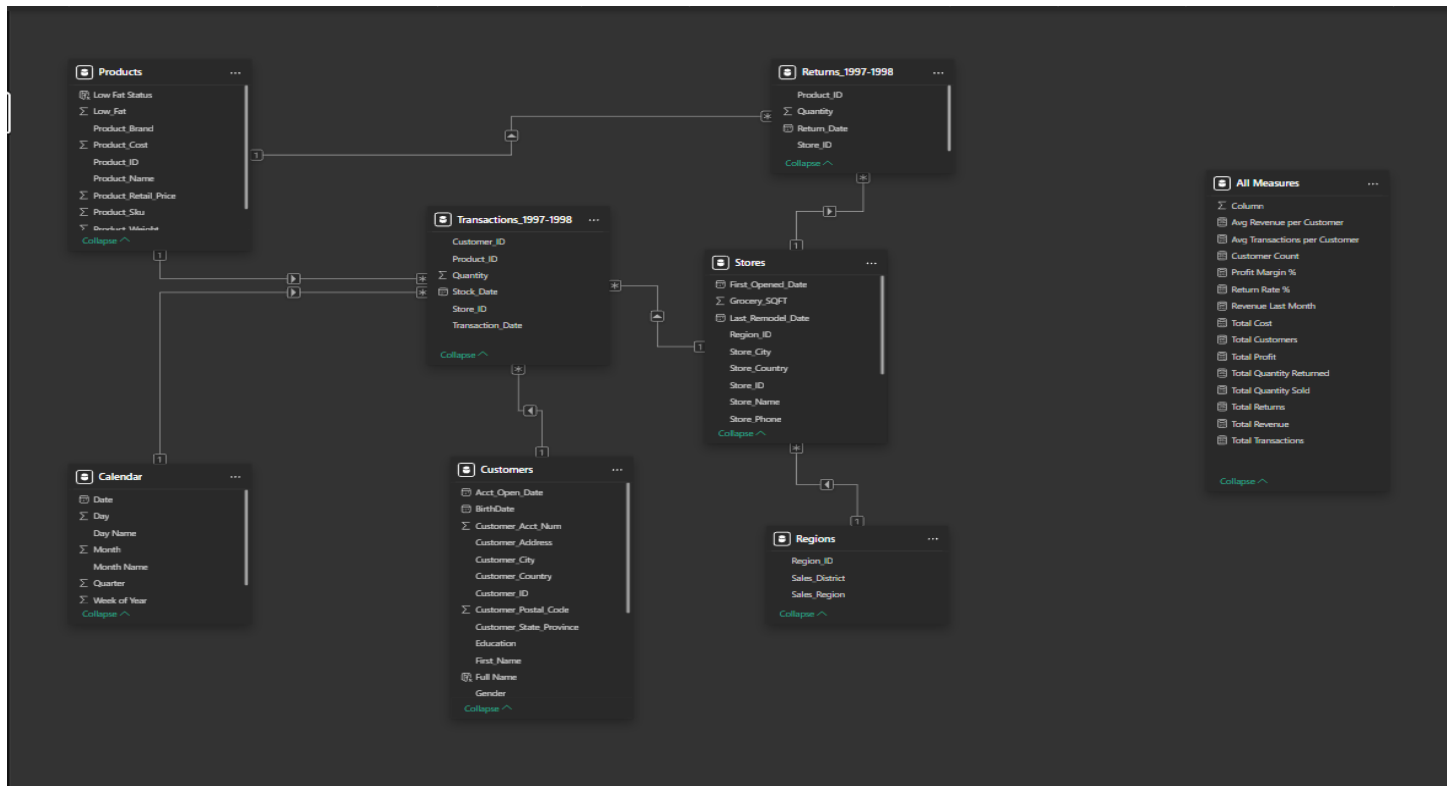
Tools Used: Power BI, DAX, Power Query.

2. Data Model & Architecture

The report is built using a **Star Schema data model** to ensure optimal performance, scalability, and accurate cross-filtering across visuals.

- **Fact Tables:** Transactions (Sales data), Returns (Return data).
- **Dimension Tables:** Customers, Products, Stores, Regions, Calendar.
- **Relationships:** One-to-Many relationships connect dimension tables to fact tables.

Data Model Diagram:

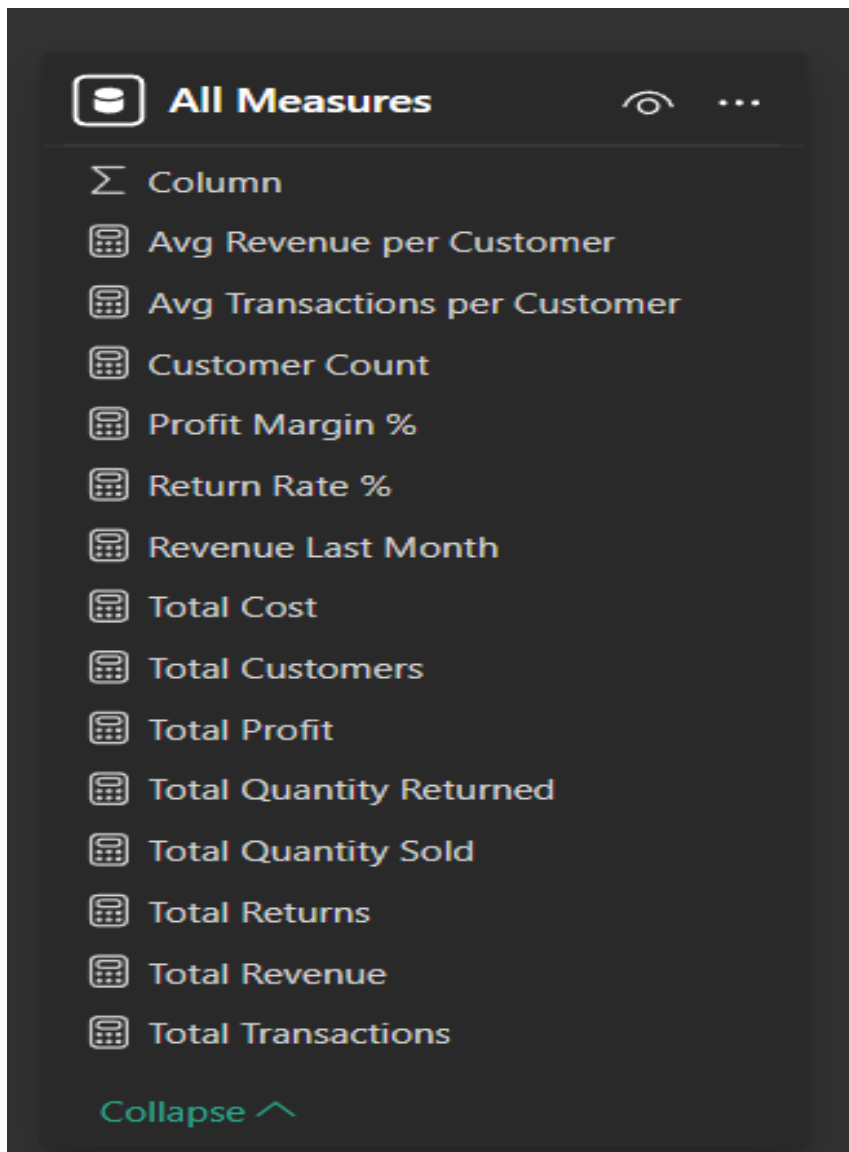


3. Key Performance Indicators (DAX Measures)

A dedicated measure table was created to house all calculations, ensuring organization and reusability. Key measures include:

- **Total Revenue:** Sum of product sales.
- **Total Profit:** Total Revenue minus Total Cost.
- **Profit Margin %:** Total Profit divided by Total Revenue.
- **Return Rate %:** Quantity Returned divided by Quantity Sold.
- **Avg Revenue per Customer:** Total Revenue divided by Unique Customers.

Measures List:



4. Report Pages & Insights

✓ Page 1: Overview

Purpose: Provide executives with a high-level overview of overall business performance.

Key Insights:

- Monthly trends highlight consistent revenue growth with seasonal profit fluctuations.
- The **USA contributes the highest share of total revenue**, positioning it as the primary growth driver.
- Identification of top-performing stores based on profit margins.

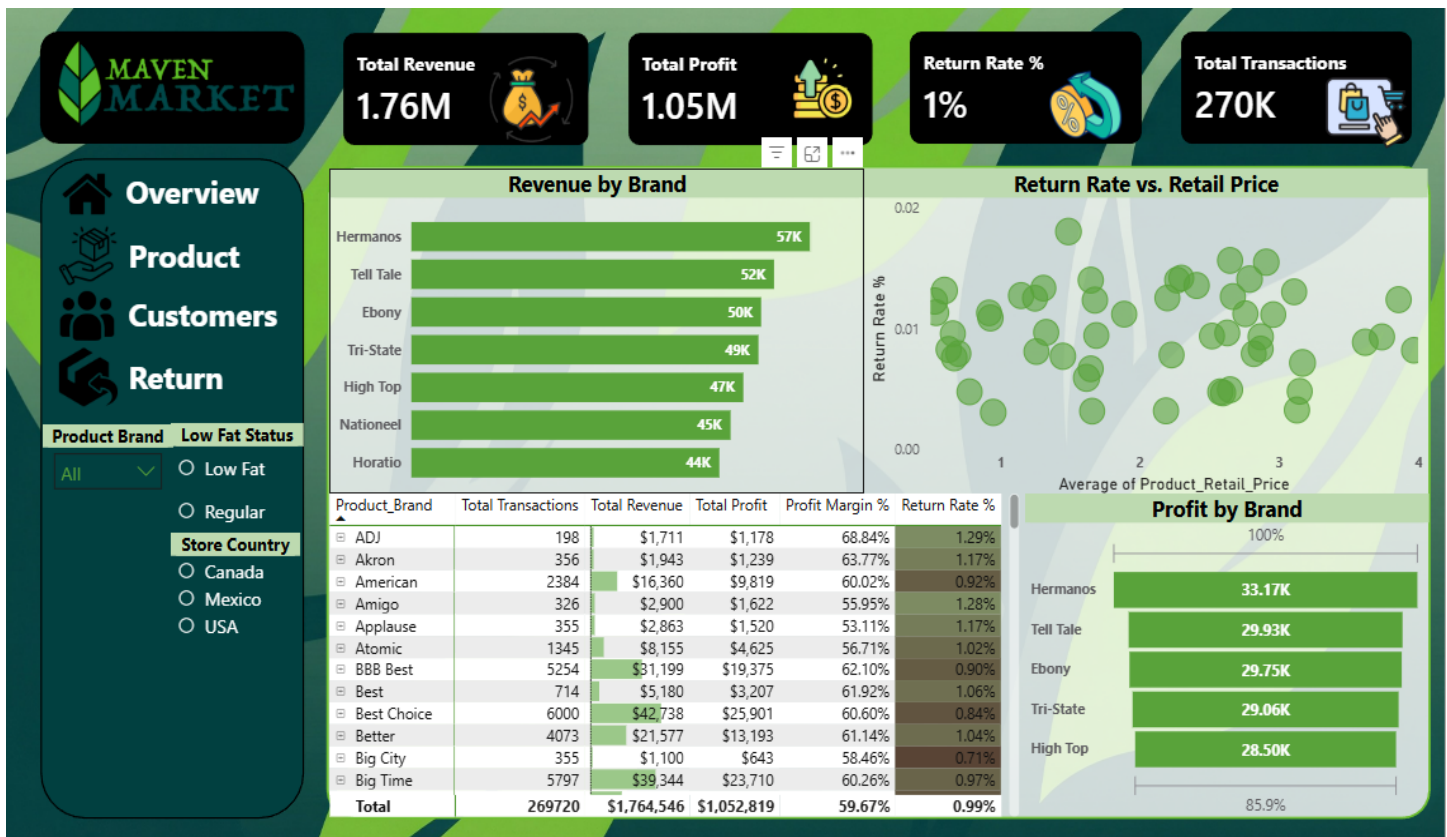


✓ Page 2: Product Analysis

Purpose: Evaluate product and brand performance to identify profitability drivers and quality concerns.

Key Insights:

- Scatter analysis of **Return Rate vs. Retail Price** reveals pricing anomalies and potential quality issues.
- Top brands by revenue and profit were identified to support supplier negotiations.
- A detailed product-level table enables granular performance analysis.



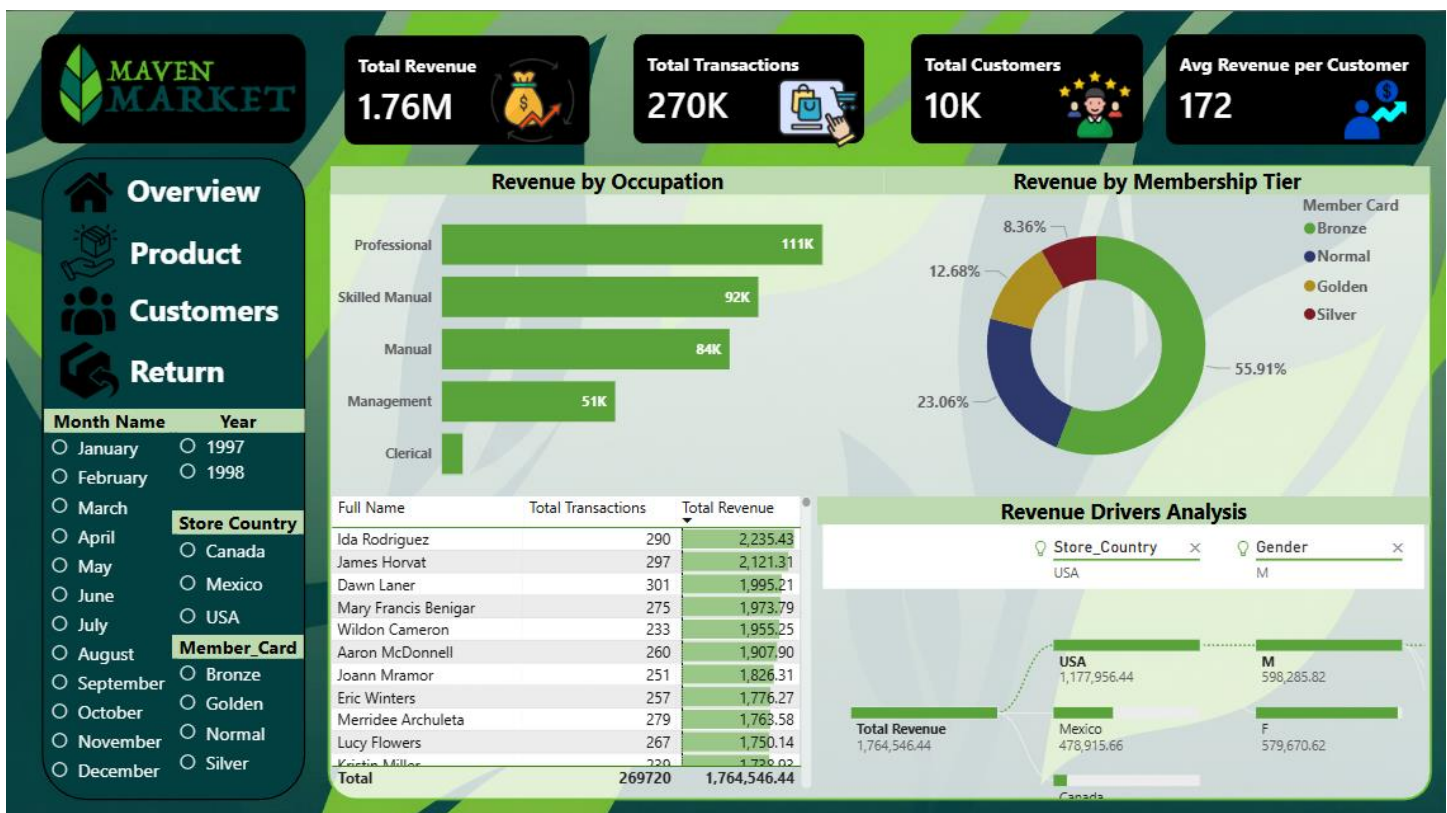
✓ Page 3: Customer Analysis

Purpose: Understand customer segments to improve targeting, retention, and loyalty strategies.

Key Insights:

- The **Professional** and **Skilled Manual** segments generate the highest revenue contribution.
- A **Decomposition Tree** enables dynamic revenue analysis by Country, Gender, and Membership status.
- A Top Customer leaderboard highlights VIP clients suitable for loyalty and reward programs.

Business Impact: These insights support personalized marketing campaigns and customer retention strategies.

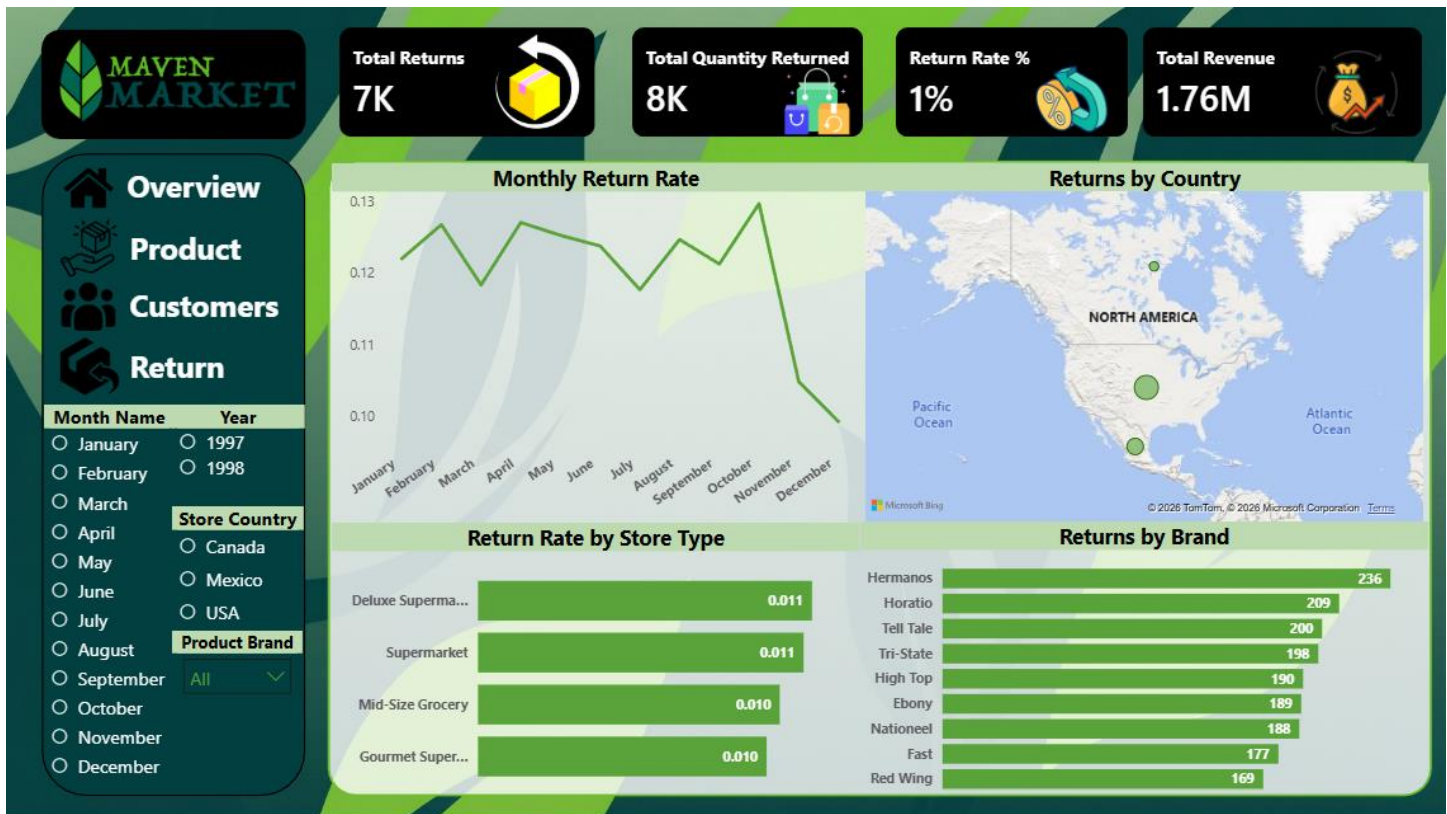


✓ Page 4: Returns Analysis

Purpose: Diagnose return behavior to minimize revenue leakage and operational inefficiencies.

Key Insights:

- Monthly return rate trends reveal identifiable return spikes.
- **Supermarket stores exhibit higher return rates** compared to other store formats.
- The top 10 brands by return volume were identified for further quality and supplier review.



5. Conclusion & Recommendations

- **Inventory Quality:** Investigate the top 10 brands with the highest return rates to address potential quality or supply chain issues.
- **Marketing Strategy:** Focus marketing and loyalty efforts on the **Professional customer segment in the USA**, as it delivers the highest revenue impact.
- **Store Operations:** Review operational processes in **Supermarket stores**, which show lower profit margins and higher return rates compared to Deluxe stores.